Lec-19

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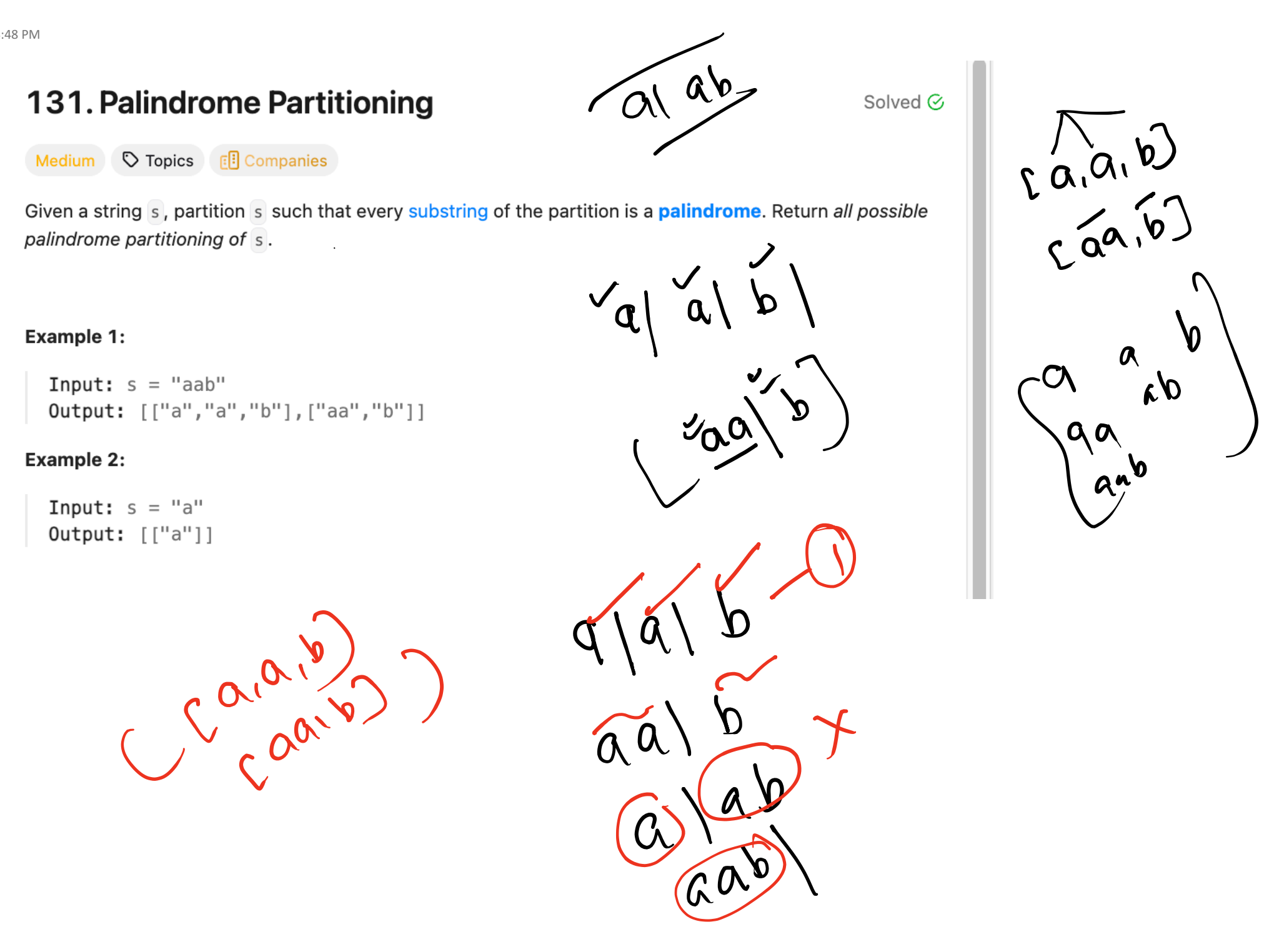


131. Palindrome Partitioning

Medium · Topics · Companies · Solved

Given a string s, partition s such that every substring of the partition is a palindrome. Return all possible palindrome partitioning of s.

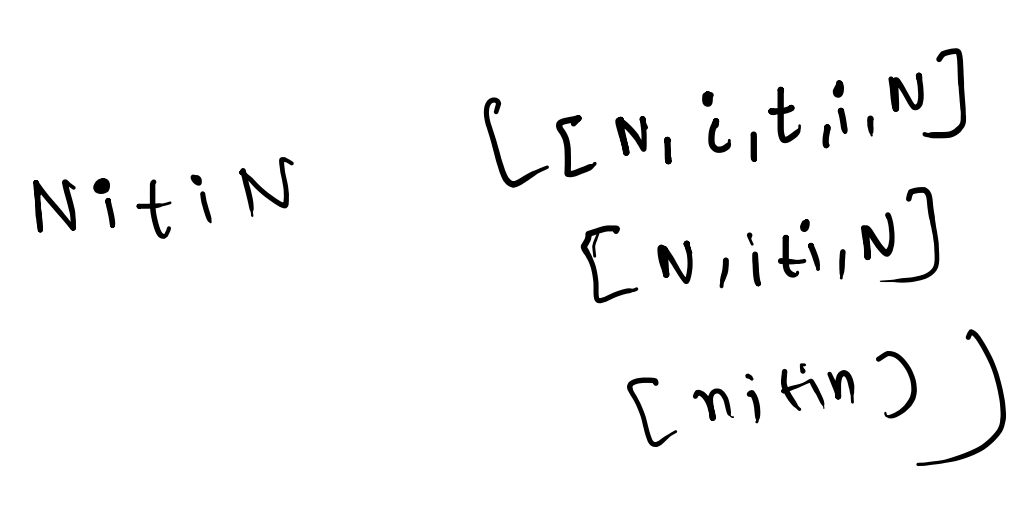
Example 1:

```
Input: s = "aab"
Output: [["a","a","b"],["aa","b"]]
```

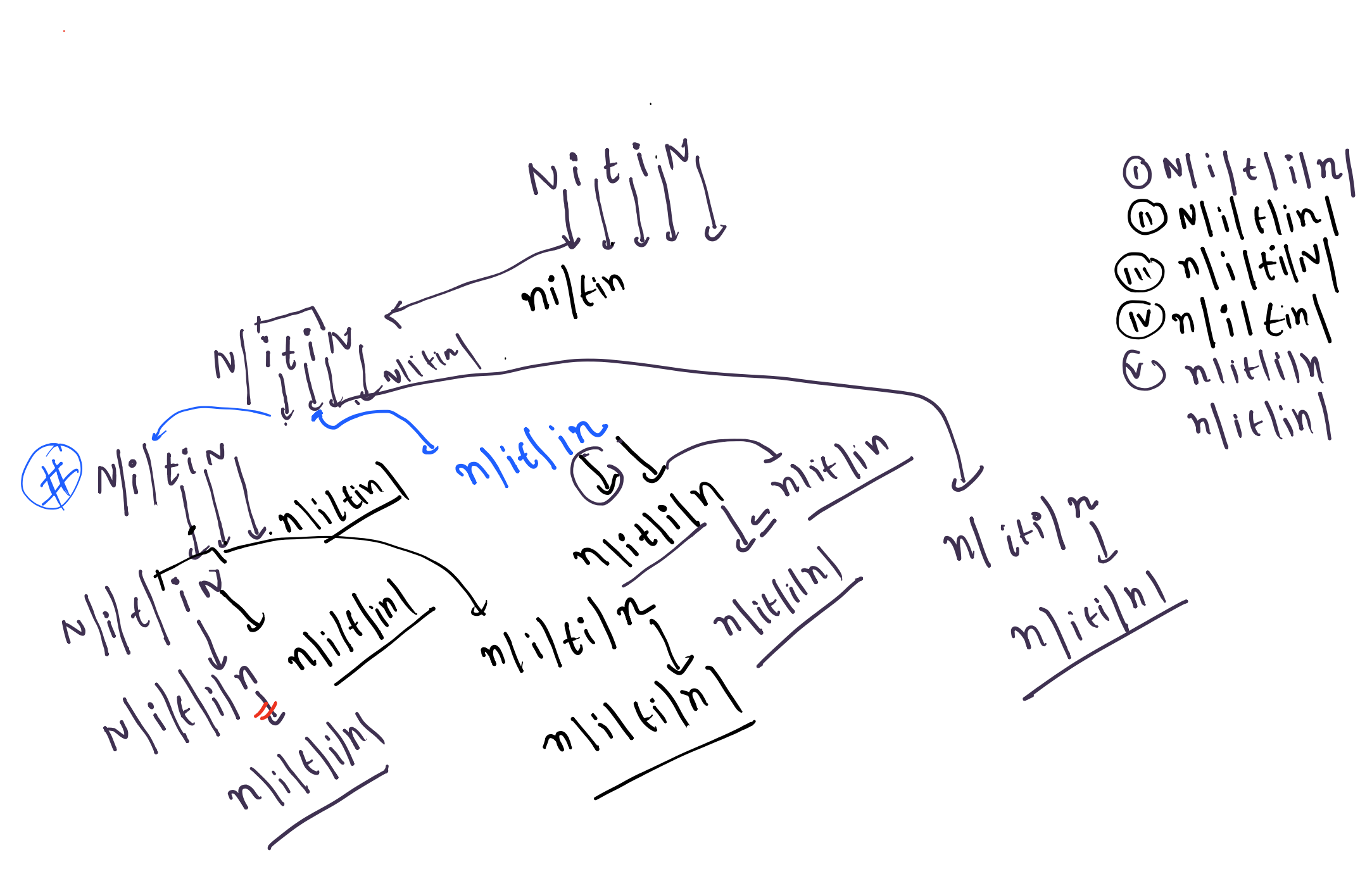
Example 2:

```
Input: s = "a"
Output: [["a"]]
```

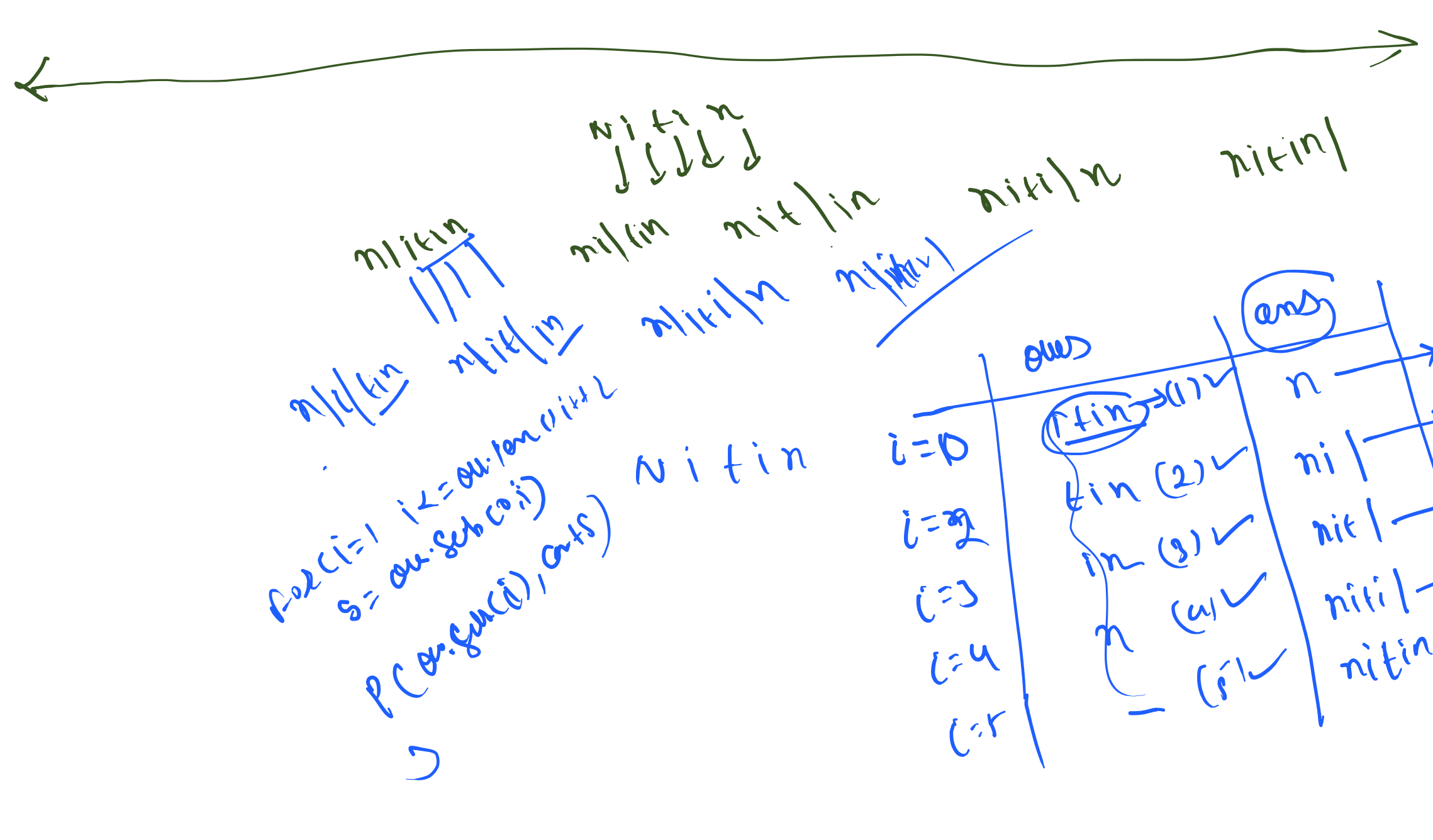
a| ab
a| a| b|
[aa| b]
a|a| b
aa| b
a| ab
aab|
[a,a,b]
[aa,b]
([a,a,b] [aab])



NitiN
[N,i,t,i,N]
[N,iti,N]
[nitin]

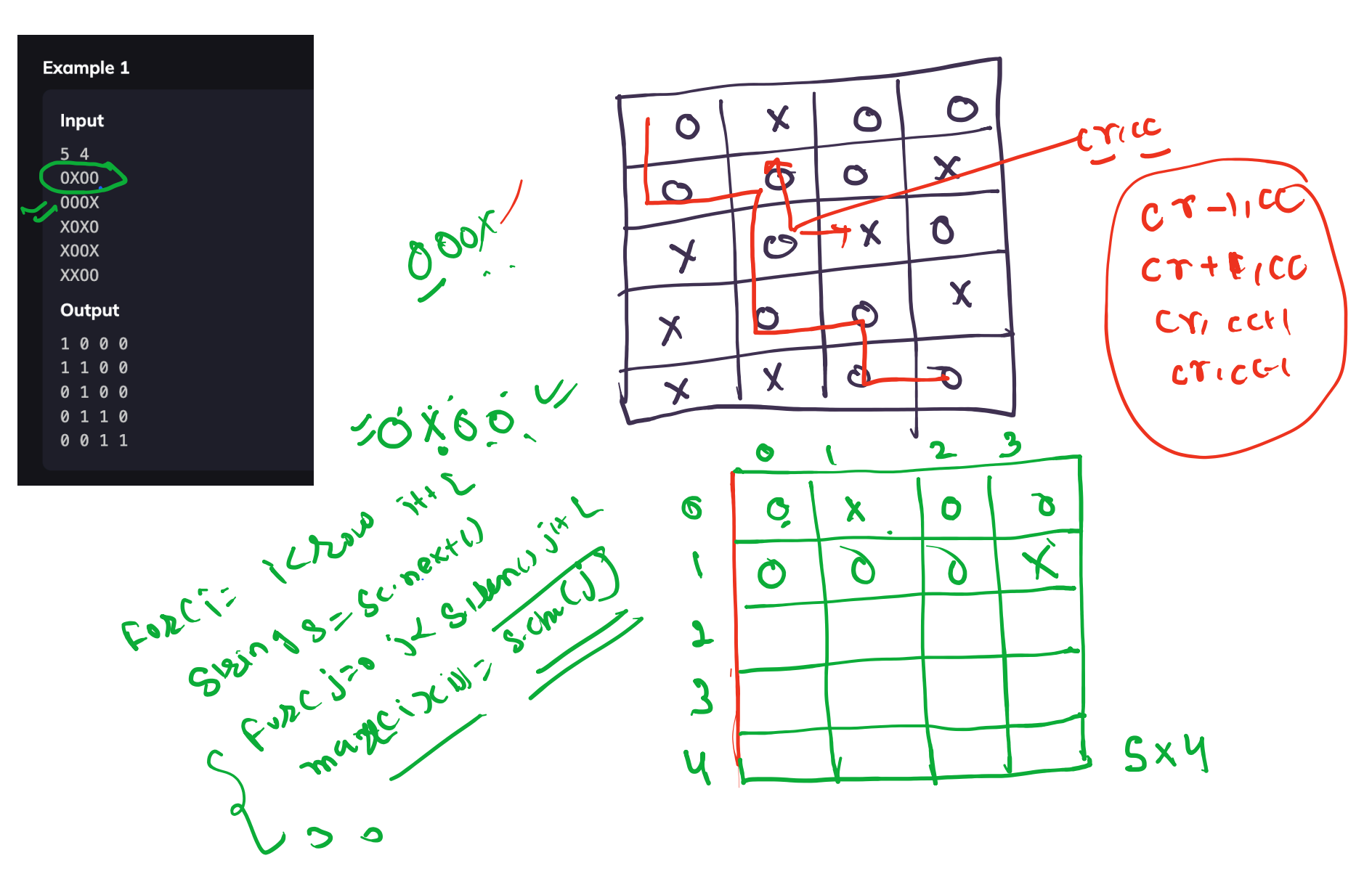


N|i|t|i|N
n|itin
N|i|t|i|n
n|i|tin
n|i|t|i|n
n|iti|n
n|itin
n|iti|n
① N|i|t|i|n|
② N|i|tin|
③ n|iti|N|
④ n|i|tin|
⑤ n|iti|n
n|itin|



nitin
n|itin
ni|tin
nit|in
niti|n
nitin|
For(i=1 i<=ques.length(); i++){
s= ques.sub(0,i)
P(ques.sub(i), ans+s)
}
Nitin
i=0
i=2
i=3
i=4
i=5
ques
ans
(itin) (1)
tin (2)
in (3)
n (4)
(5)
n
ni|
nit|
niti|
nitin|
(0,1)
(0,2)
0,3)
0,4
0,5

n|i|t|i|n|
n|i|t|in|
n|i|ti|n|
n|i|tin|
n|it|i|n|
n|it|in|
n|iti|n|
n|itin|
ni|t|i|n|
ni|t|in|
ni|ti|n|
ni|tin|
nit|i|n|
nit|in|
niti|n|
nitin|



Example 1

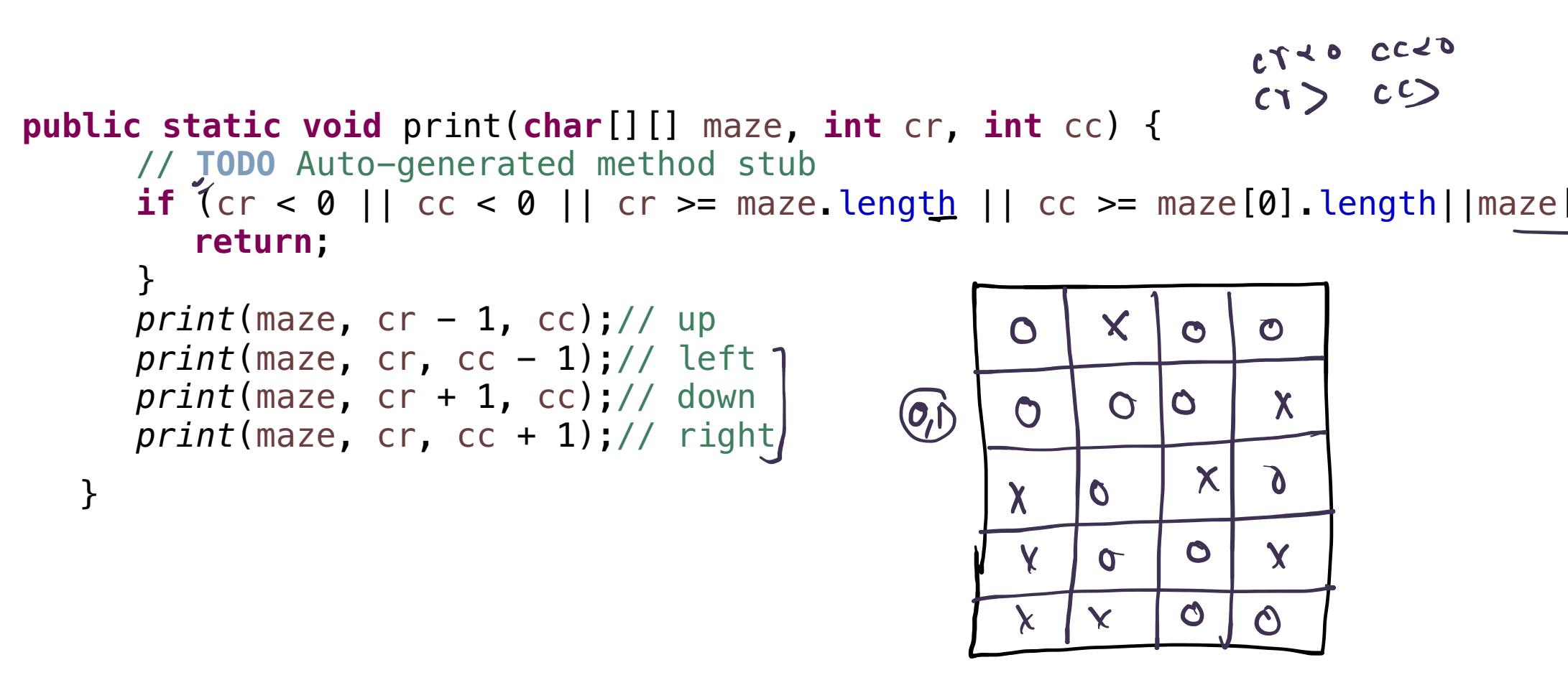
Input

```
5 4
OXOO
OOOX
XOXO
XOOX
XXOO
```

Output

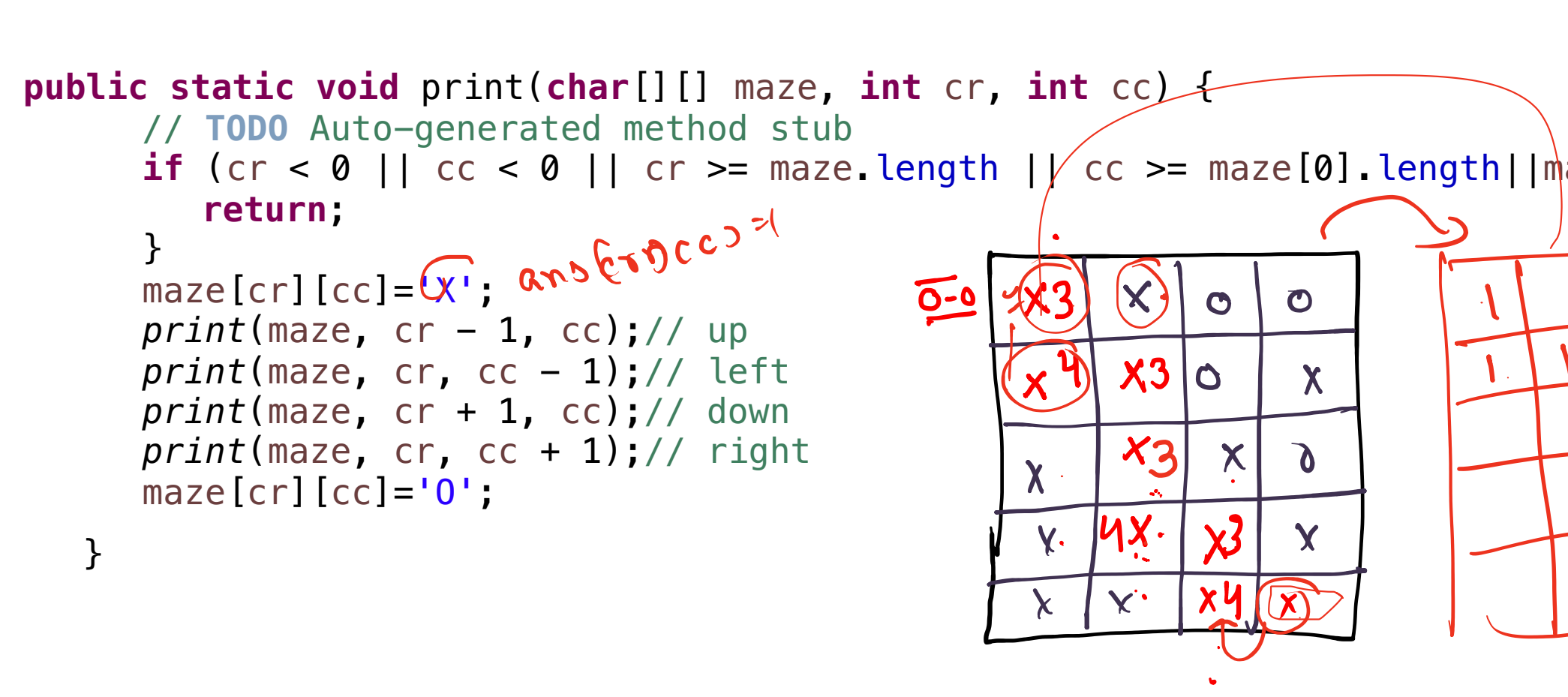
```
1 0 0 0
1 1 0 0
0 1 0 0
0 1 1 0
0 0 1 1
```

OOOX
=OXOO
For(i=1 i<row; i++){
String s=sc.next()
For(j=0 j<s.length(); j++){
maze[i][j]= s.charAt(j)
}
}
cr,cc
cr-1,cc
cr+1,cc
cr,cc+1
cr,cc-1
0 1 2 3
5x4



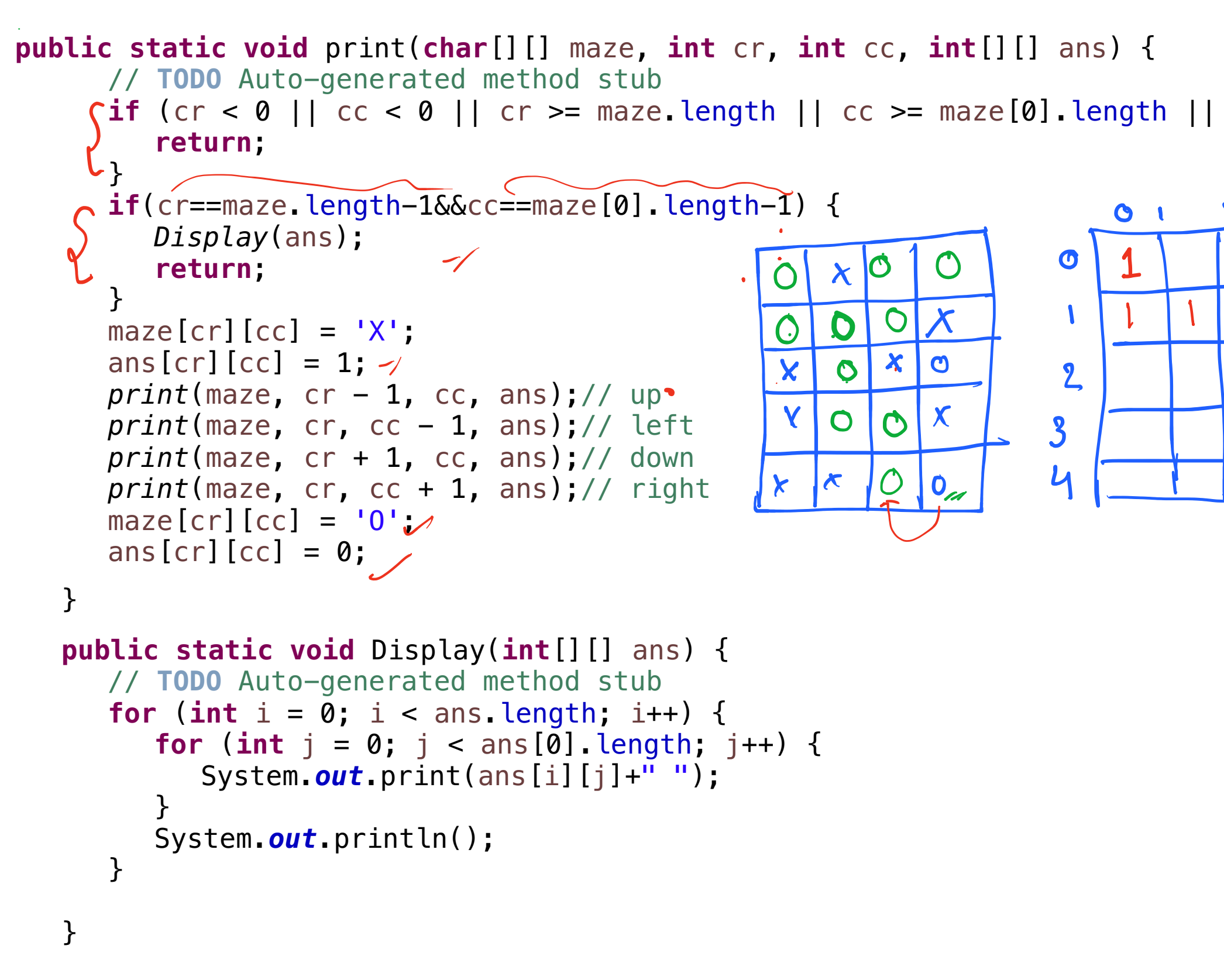
```
public static void print(char[][] maze, int cr, int cc) {
    // TODO Auto-generated method stub
    if (cr < 0 || cc < 0 || cr >= maze.length || cc >= maze[0].length||maze[cr][cc]=='X') {
        return;
    }
    print(maze, cr - 1, cc);// up
    print(maze, cr, cc - 1);// left
    print(maze, cr + 1, cc);// down
    print(maze, cr, cc + 1);// right
}
```

cr<0 cc<0
cr> cc>
0,0



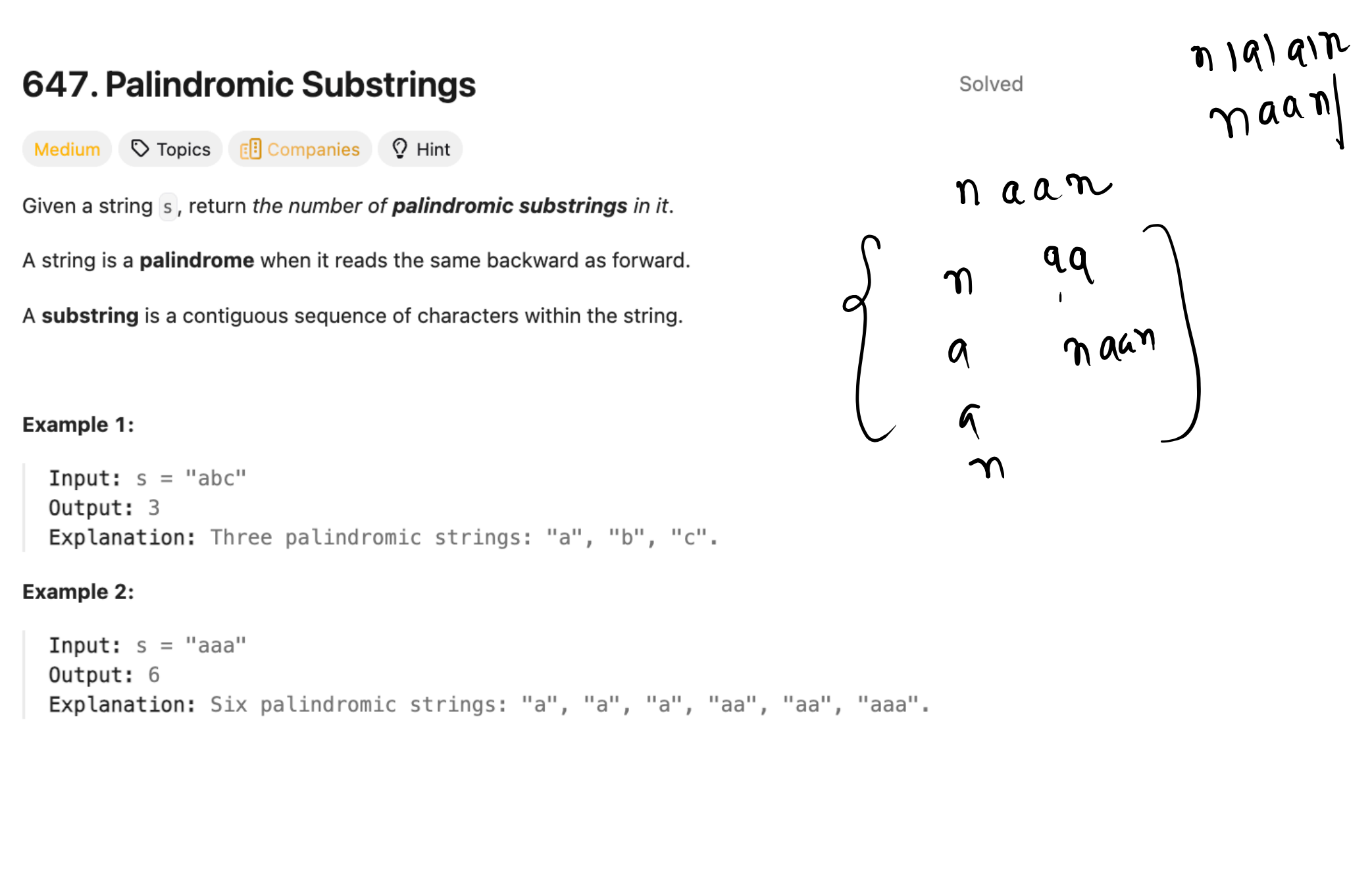
```
public static void print(char[][] maze, int cr, int cc) {
    // TODO Auto-generated method stub
    if (cr < 0 || cc < 0 || cr >= maze.length || cc >= maze[0].length||maze[cr][cc]=='X') {
        return;
    }
    maze[cr][cc]='X';
    print(maze, cr - 1, cc);// up
    print(maze, cr, cc - 1);// left
    print(maze, cr + 1, cc);// down
    print(maze, cr, cc + 1);// right
    maze[cr][cc]='O';
}
```

ans[cr][cc]=1
0-0
X3
X4
4X



```
public static void print(char[][] maze, int cr, int cc, int[][] ans) {
    // TODO Auto-generated method stub
    if (cr < 0 || cc < 0 || cr >= maze.length || cc >= maze[0].length || maze[cr][cc] == 'X') {
        return;
    }
    if(cr==maze.length-1&&cc==maze[0].length-1) {
        Display(ans);
        return;
    }
    maze[cr][cc] = 'X';
    ans[cr][cc] = 1;
    print(maze, cr - 1, cc, ans);// up
    print(maze, cr, cc - 1, ans);// left
    print(maze, cr + 1, cc, ans);// down
    print(maze, cr, cc + 1, ans);// right
    maze[cr][cc] = 'O';
    ans[cr][cc] = 0;
}

public static void Display(int[][] ans) {
    // TODO Auto-generated method stub
    for (int i = 0; i < ans.length; i++) {
        for (int j = 0; j < ans[0].length; j++) {
            System.out.print(ans[i][j]+" ");
        }
        System.out.println();
    }
}
```



647. Palindromic Substrings

Medium · Topics · Companies · Hint · Solved

Given a string s, return the number of **palindromic substrings** in it.

A string is a **palindrome** when it reads the same backward as forward.

A **substring** is a contiguous sequence of characters within the string.

Example 1:

```
Input: s = "abc"
Output: 3
Explanation: Three palindromic strings: "a", "b", "c".
```

Example 2:

```
Input: s = "aaa"
Output: 6
Explanation: Six palindromic strings: "a", "a", "a", "aa", "aa", "aaa".
```

n|a|a|n
naan|
naan
n aa
a naan
a
n

Concept

① Count Palindrome Substring

(ii) Print largest len PS

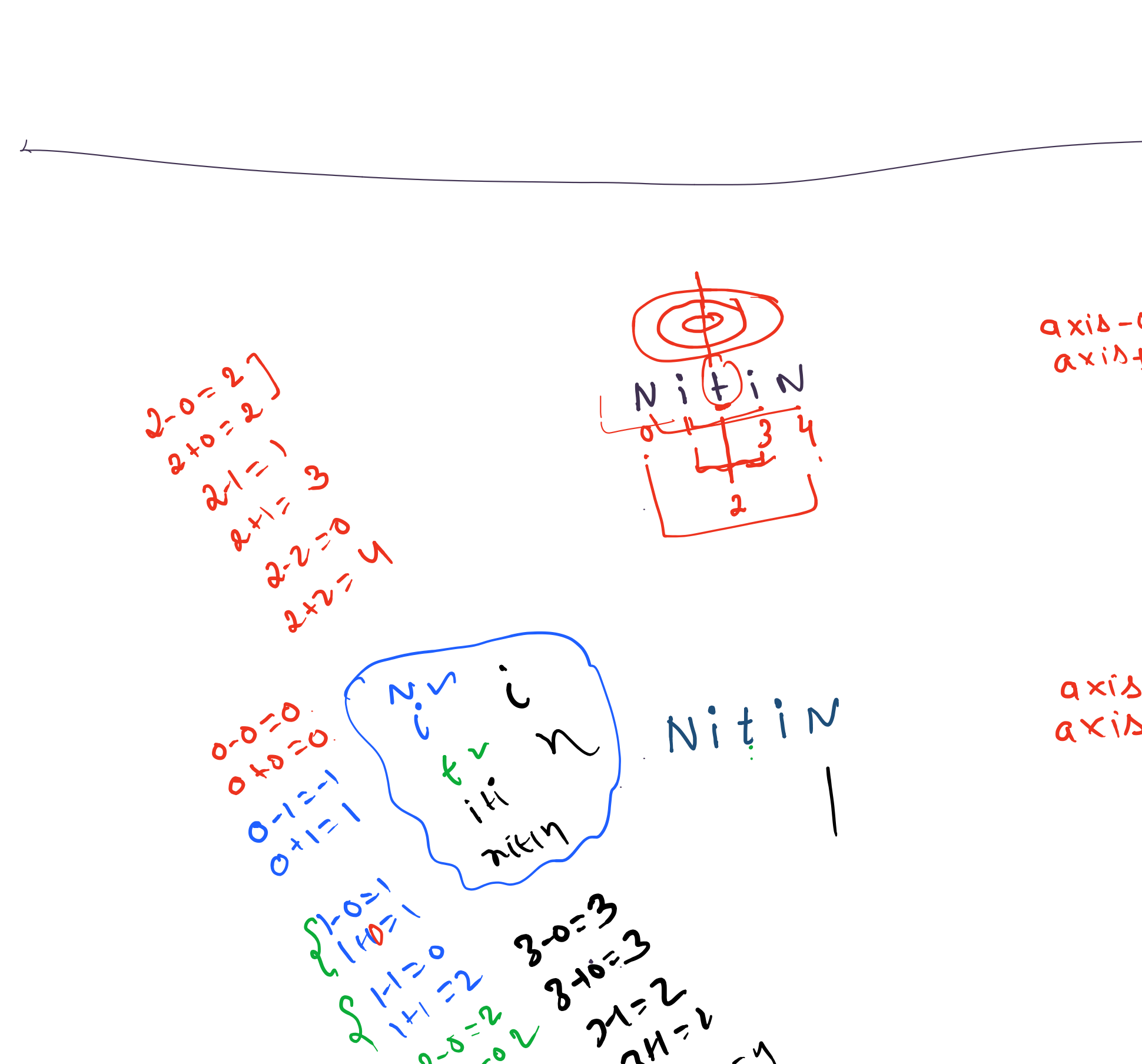
(iii) Print len of PLS

(iv) Count odd len PLS

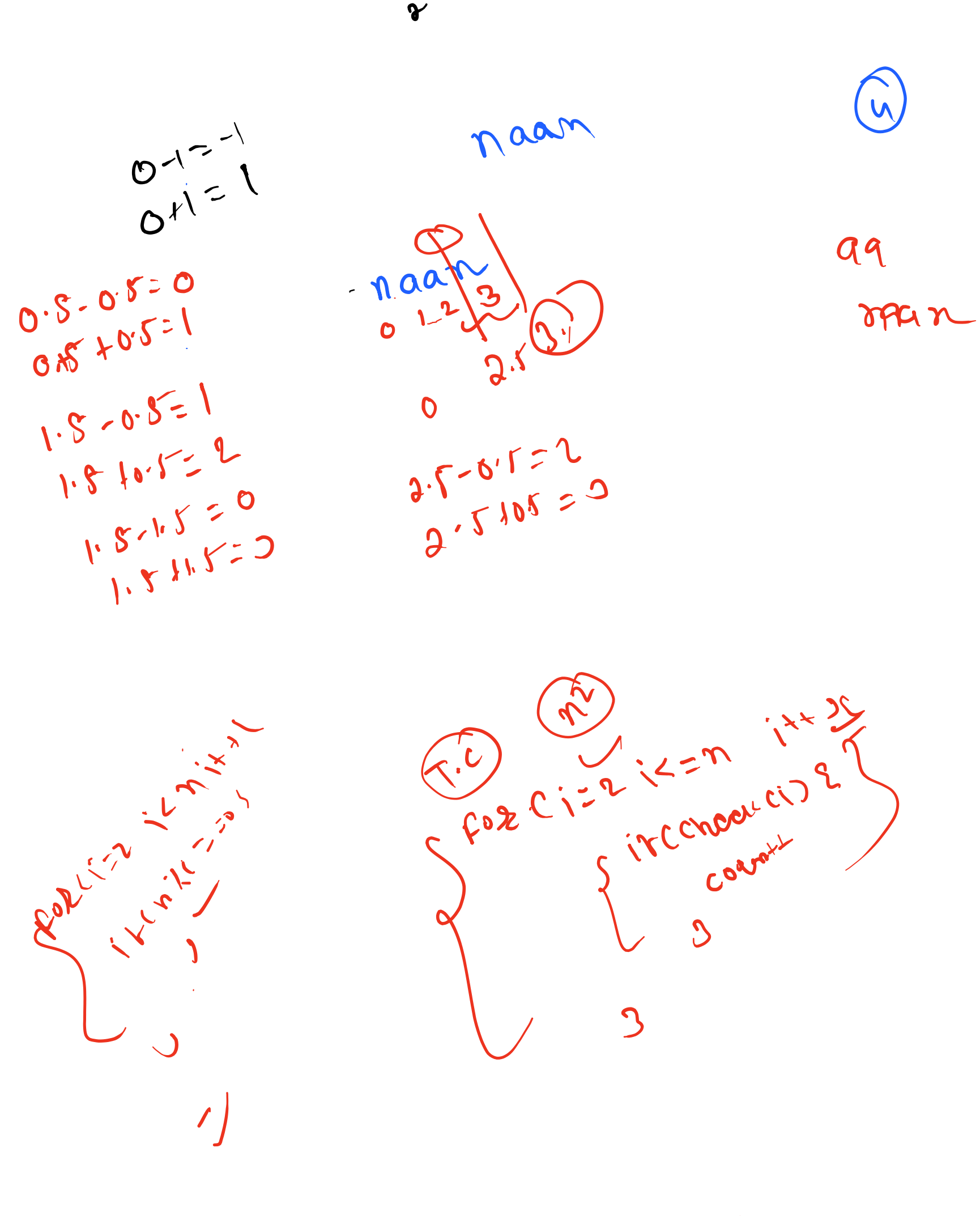
(v) Count both even len PLS

(vi) Largest even PLS

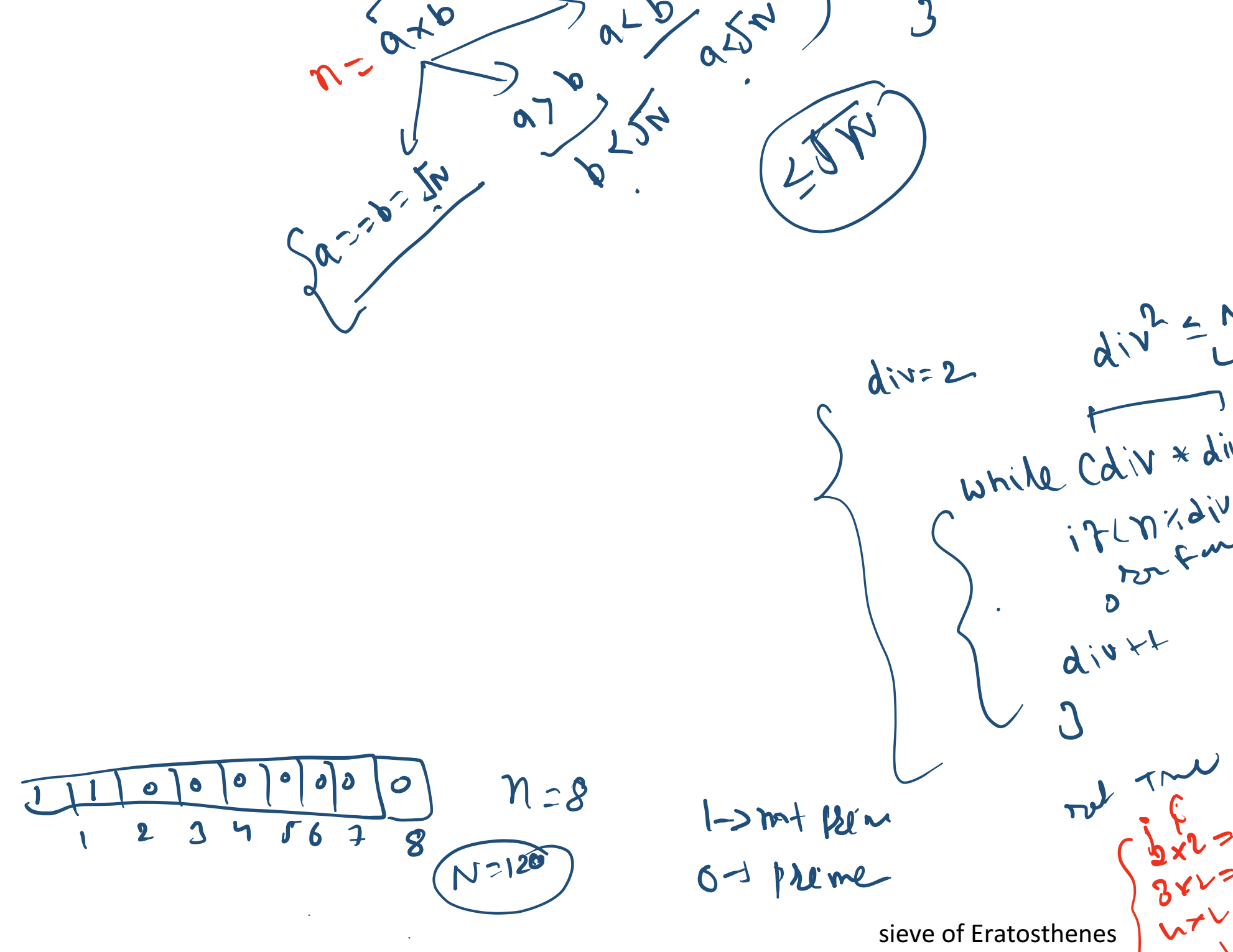
od PLS



NitiN
2-0=2
2+0=2
2-1=1
2+1=3
2-2=0
2+2=4
axis-orbit
axis+orbit
t
iti
nitin
0-0=0
0+0=0
0-1=-1
0+1=1
i
n
iti
nitin
1-0=1
1+0=1
1-1=0
1+1=2
2-0=2
2+0=2
2-1=1
2+1=3
2-2=0
2+2=4
3-0=3
3+0=3
3-1=2
3+1=4
4-0=4
4+0=4
Nitin
axis-orbit
axis+orbit



0-1=-1
0+1=1
naan
0 1 2 3
0.5-0.5=0
0.5+0.5=1
1.5-0.5=1
1.5+0.5=2
1.5-1.5=0
1.5+1.5=3
2.5-0.5=2
2.5+0.5=3
aa
naan
for(i=2 i<n i++)
if(checkprime(i)){
count++
}
T.C
n^2
$n^2=(10^8)^2=10^{16}$
for(i=2 i<=n i++){
if(checkprime(i)){
}
O(n)
$\sqrt{n}$
$a^2=n$
$b^2=n$
n=a×b
a<b
a≤√n
a>b
b<√n
a==b=√n
≤√n
div=2
$div^2 \le N$
div ≤ √N
while(div*div<=N){
if(n%div==0)
ret false
div++
}
ret True

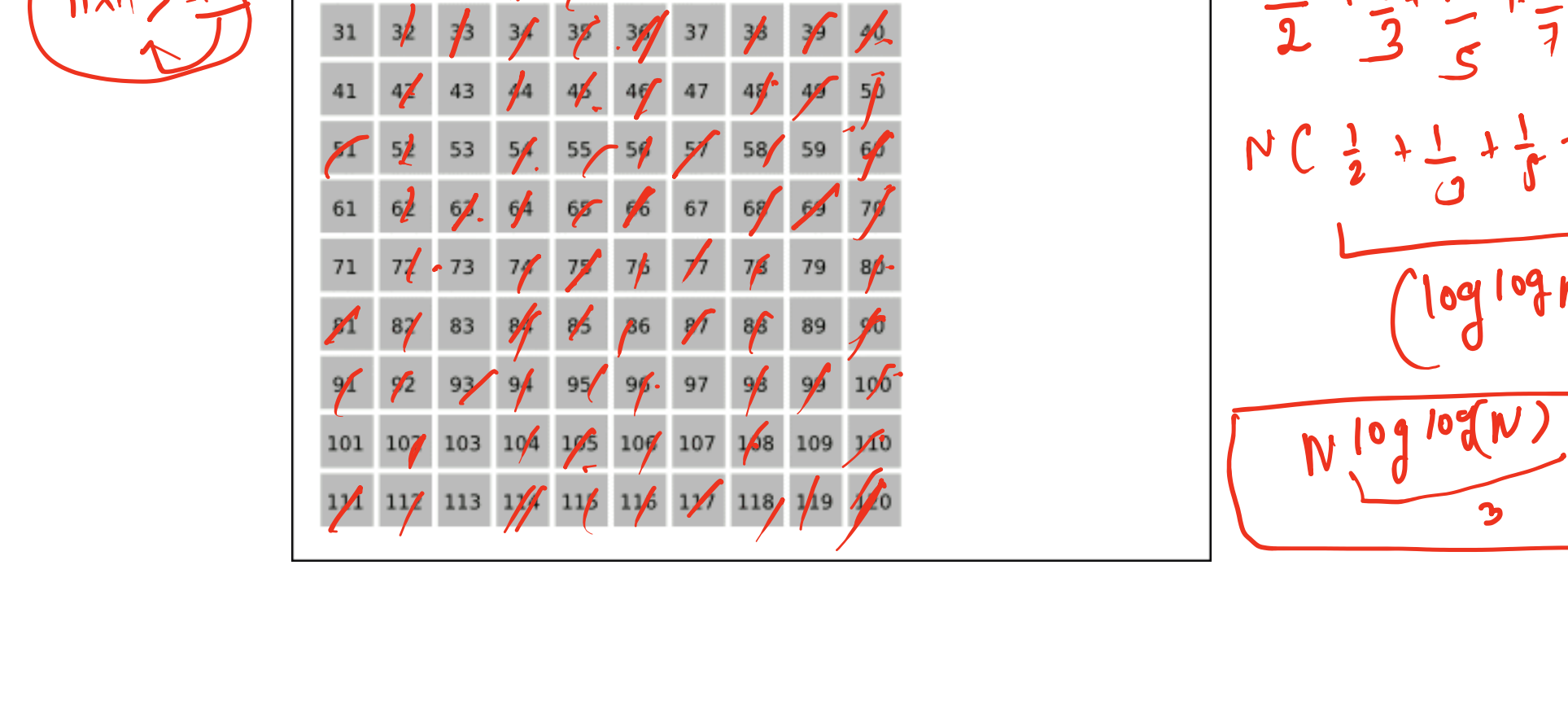


1 1 0 0 0 0 0 0 0
1 2 3 4 5 6 7 8
n=8
N=120
1->not prime
0->prime
11x11>120
i=2 i=3 i=5 i=7
N/2 + N/3 + N/5 + N/7
N(1/2 + 1/3 + 1/5 + 1/7 ...)
(log log N)
N log log(N)
N=1024
2^10
log log 2^10
log 10
=3...

sieve of Eratosthenes

Prime numbers

	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100
101	102	103	104	105	106	107	108	109	110
111	112	113	114	115	116	117	118	119	120



```
public static int Sieve(int n) {
    int[] prime = new int[n + 1];
    prime[0] = 1;// not prime
    prime[1] = 1;// not prime
    for (int i = 2; i * i < prime.length; i++) {
        if (prime[i] == 0) {
            for (int j = 2; j * i < prime.length; j++) {
                prime[j * i] = 1;
            }
        }
    }
}
```

N=20
i=3
5x5<20
4x4<20
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20